Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester VI)
Artificial Intelligence Practical

Practical – Task I

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1. Problem Statement

The objective of this practical is to implement various Artificial Intelligence search algorithms to determine the most efficient route between **MVLU College (Source)** and **Bamandaya Pada (Home)** located in Andheri East, Mumbai.

The following search algorithms are implemented and compared:

- Breadth-First Search (BFS)
- Depth-First Search (DFS)
- Uniform Cost Search (UCS)
- Greedy Best-First Search
- A* Search

Each algorithm is evaluated based on:

- Path Found
- Total Distance (Cost)
- Number of Nodes Explored

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2. Graph Diagram

The route between MVLU College and Bamandaya Pada is represented as a weighted undirected graph.

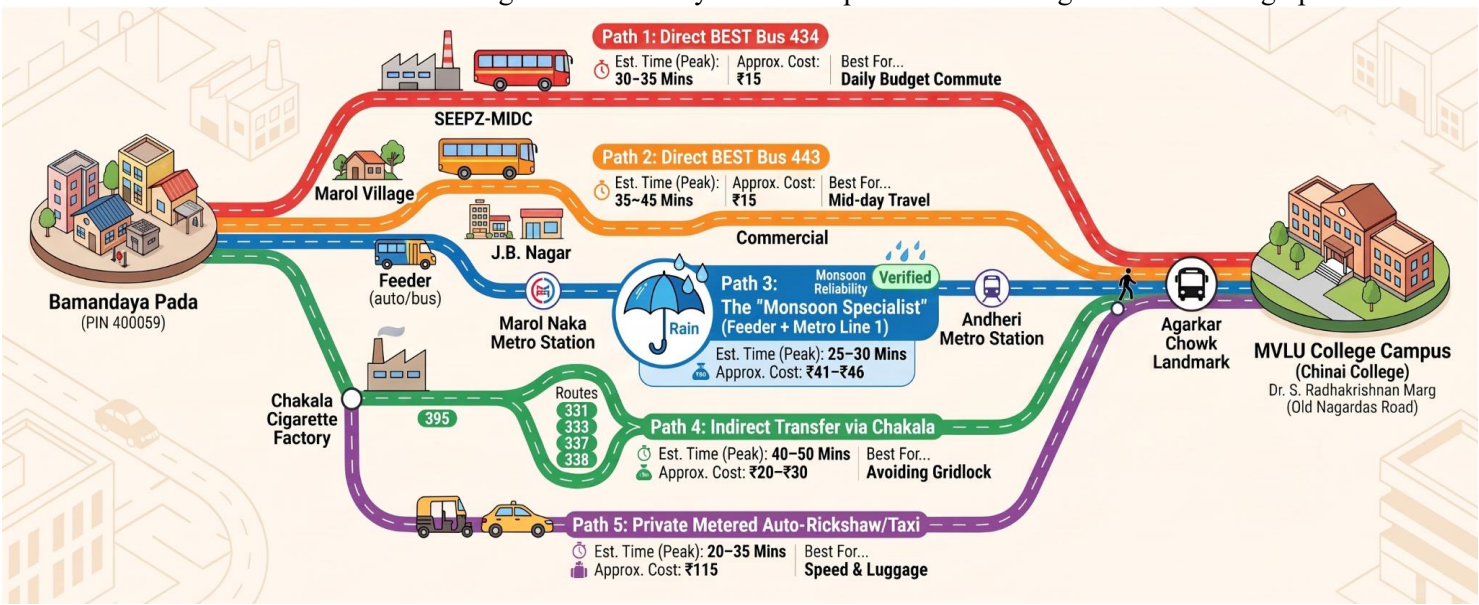

Figure 1: InfoGraphical representation of road networks between MVLU College and Bamandaya Pada.

3. Edge Cost Table

Source Node	Destination Node	Edge Cost / Distance (g(n))
MVLU College	Agarkar Chowk	0.3
Agarkar Chowk	Vishal Hall	0.4
Vishal Hall	Chakala	0.5
Chakala	Orkay Mill	0.5
Orkay Mill	C-Cross / National Test House	0.4
C-Cross	SEEPZ Bus Station	0.5
SEEPZ Bus Station	SEEPZ Village	0.5
SEEPZ Village	Aarey JVLR Metro	0.7
Aarey JVLR Metro	Milind Nagar	1.1
Milind Nagar	Bamandaya Pada	0.6
SEEPZ Bus Station	MIDC Police Station	0.8
MIDC Police Station	Gautam Nagar	0.5
Gautam Nagar	Nelco	0.6
Nelco	Chakala	0.7
MIDC Police Station	Marol Depot	0.8
Marol Depot	Nelco	0.7
IES School	Milind Nagar	0.4
IES School	SEEPZ Bus Station	0.6

4. Heuristic Table

(Straight-line estimated distance to **Bamandaya Pada**)

Node Name	Heuristic Value h(n) (km)
MVLU College	6.8

Agarkar Chowk	6.4
Vishal Hall	6.0
Chakala	5.5
Orkay Mill	5.0
C-Cross	4.6
SEEPZ Bus Station	4.1
SEEPZ Village	3.5
MIDC Police Station	3.0
Gautam Nagar	2.6
Marol Depot	2.4
Nelco	2.0
Aarey JVLR Metro	1.3
IES School	0.9
Milind Nagar	0.5
Bamandaya Pada	0
Durga Nagar	0.0

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5. Brief Description of Search Algorithms

1. **Breadth-First Search (BFS)**
   Breadth-First Search explores all neighboring nodes level by level using a **Queue (FIFO)**. It guarantees the shortest path in terms of the **number of edges**, but it does not consider edge costs.
2. **Depth-First Search (DFS)**
   Depth-First Search explores one branch completely before backtracking. It uses a **Stack (LIFO)** and may not always find the shortest or least-cost path.
3. **Uniform Cost Search (UCS)**
   Uniform Cost Search expands the node having the **lowest cumulative path cost**. It always finds the least-cost path when all edge costs are non-negative.
4. **Greedy Best-First Search**
   Greedy Best-First Search selects the node with the smallest heuristic value **h(n)**. It is usually fast but does not always produce the optimal path.
5. **A* Search**

A* Search combines the actual path cost **g(n)** and the heuristic estimate **h(n)** to compute **f(n) = g(n) + h(n)**. It is widely used in GPS navigation because it efficiently finds optimal paths when the heuristic is admissible.

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6. Python Source Code & Program Outputs

6.1 Breadth-First Search (BFS)

Algorithm Description

Breadth-First Search (BFS) is an uninformed search algorithm that explores all neighboring nodes before moving to the next level. It uses a **First-In First-Out (FIFO) Queue**.

BFS guarantees the path with the **minimum number of edges (hops)** but does **not** guarantee the minimum travel cost when edge weights are different.

Python Program (BFS)

SOURCE:-

```
from collections import deque
# Graph (Adjacency List with Edge Costs)
graph = {
   "MVLU College": [("Agarkar Chowk",0.3)],
   "Agarkar Chowk":[("MVLU College",0.3),
                ("Vishal Hall",0.4)],
   "Vishal Hall":[("Agarkar Chowk",0.4),
               ("Chakala",0.5)],
   "Chakala":[("Vishal Hall",0.5),
           ("Orkay Mill",0.5),
           ("Nelco",0.7)],
   "Orkay Mill":[("Chakala",0.5),
              ("C-Cross",0.4)],
   "C-Cross":[("Orkay Mill",0.4),
           ("SEEPZ Bus Station",0.5)],
   "SEEPZ Bus Station":[("C-Cross",0.5),
                    ("SEEPZ Village",0.5),
                    ("MIDC Police Station",0.8),
                    ("IES School",0.6)],
   "SEEPZ Village":[("SEEPZ Bus Station",0.5),
                ("Aarey JVLR Metro",0.7)],
   "MIDC Police Station":[("SEEPZ Bus Station",0.8),
                      ("Gautam Nagar",0.5),
                      ("Marol Depot",0.8)],
   "Gautam Nagar":[("MIDC Police Station",0.5),
                ("Nelco",0.6)],
   "Marol Depot":[("MIDC Police Station",0.8),
                ("Nelco",0.7)],
   "Nelco":[("Marol Depot",0.7),
          ("Gautam Nagar",0.6),
          ("Chakala",0.7)],
   "Aarey JVLR Metro":[("SEEPZ Village",0.7),
                    ("Milind Nagar",1.1)],
   "IES School":[("Milind Nagar",0.4),
              ("SEEPZ Bus Station",0.6)],
```

```
    "Milind Nagar":[("IES School",0.4),
                    ("Aarey JVLR Metro",1.1),
                    ("Bamandaya Pada",0.6)],
    "Bamandaya Pada":[("Milind Nagar",0.6)]
}
def path_cost(path):
    cost = 0
    for i in range(len(path)-1):
        for node,w in graph[path[i]]:
            if node==path[i+1]:
                cost += w
                break
    return round(cost,2)
def bfs(start,goal):
    queue = deque([[start]])
    visited = set()
    explored = 0
    while queue:
        path = queue.popleft()
        node = path[-1]
        if node not in visited:
            visited.add(node)
            explored += 1
            if node==goal:
                return path,path_cost(path),explored
            for neighbour,_ in graph[node]:
                if neighbour not in visited:
                    queue.append(path+[neighbour])
start = "MVLU College"
goal = "Bamandaya Pada"
path,cost,nodes = bfs(start,goal)
print("\n========== BFS OUTPUT ==========")
print("Source :",start)
print("Destination :",goal)
print("Path :", " -> ".join(path))
print("Total Cost :",cost,"km")
print("Nodes Explored :",nodes)
print("Lavkush T048")
```

OUTPUT

```
========== BFS OUTPUT ==========
Source : MVLU College
Destination : Bamandaya Pada
Path : MVLU College -> Agarkar Chowk -> Vishal Hall -> Chakala -> Orkay Mill -> C-Cross -> SEEPZ
Bus Station -> IES School -> Milind Nagar -> Bamandaya Pada
Total Cost : 4.2 km
Nodes Explored : 16
Lavkush T048
```

Explanation

BFS starts from **MVLU College** and explores all neighboring nodes level by level.

Since BFS considers only the number of edges and ignores edge costs, it finds the first valid path that reaches **Bamandaya Pada**. Although the path is valid, it may not always be the least-cost route when compared with UCS or A* Search.

6.2 Depth-First Search (DFS)
Algorithm Description

Depth-First Search (DFS) is an uninformed search algorithm that explores one path as deeply as possible before backtracking. It uses a **Last-In First-Out (LIFO) Stack**.
Unlike BFS, DFS does **not** guarantee the shortest or least-cost path. The result depends on the order in which neighboring nodes are explored.

Python Program (DFS)
SOURCE:-

```
# Graph (Adjacency List with Edge Costs)
graph = {
   "MVLU College":[("Agarkar Chowk",0.3)],
   "Agarkar Chowk":[("MVLU College",0.3),
                   ("Vishal Hall",0.4)],
   "Vishal Hall":[("Agarkar Chowk",0.4),
                  ("Chakala",0.5)],
   "Chakala":[("Vishal Hall",0.5),
              ("Orkay Mill",0.5),
              ("Nelco",0.7)],
   "Orkay Mill":[("Chakala",0.5),
                 ("C-Cross",0.4)],
   "C-Cross":[("Orkay Mill",0.4),
              ("SEEPZ Bus Station",0.5)],
   "SEEPZ Bus Station":[("C-Cross",0.5),
                        ("SEEPZ Village",0.5),
                        ("MIDC Police Station",0.8),
                        ("IES School",0.6)],
   "SEEPZ Village":[("SEEPZ Bus Station",0.5),
                    ("Aarey JVLR Metro",0.7)],
   "MIDC Police Station":[("SEEPZ Bus Station",0.8),
                          ("Gautam Nagar",0.5),
                          ("Marol Depot",0.8)],
   "Gautam Nagar":[("MIDC Police Station",0.5),
                   ("Nelco",0.6)],
   "Marol Depot":[("MIDC Police Station",0.8),
                  ("Nelco",0.7)],
   "Nelco":[("Marol Depot",0.7),
            ("Gautam Nagar",0.6),
            ("Chakala",0.7)],
   "Aarey JVLR Metro":[("SEEPZ Village",0.7),
                       ("Milind Nagar",1.1)],
   "IES School":[("Milind Nagar",0.4),
                 ("SEEPZ Bus Station",0.6)],
   "Milind Nagar":[("IES School",0.4),
                   ("Aarey JVLR Metro",1.1),
                   ("Bamandaya Pada",0.6)],
   "Bamandaya Pada":[("Milind Nagar",0.6)]
}
def path_cost(path):
   cost = 0
   for i in range(len(path)-1):
      for neighbour, weight in graph[path[i]]:
```

```
            if neighbour == path[i+1]:
                cost += weight
                break
    return round(cost,2)
def dfs(start, goal):
    stack = [[start]]
    visited = set()
    explored = 0
    while stack:
        path = stack.pop()
        node = path[-1]
        if node not in visited:
            visited.add(node)
            explored += 1
            if node == goal:
                return path, path_cost(path), explored
            # Reverse order for consistent traversal
            for neighbour, _ in reversed(graph[node]):
                if neighbour not in visited:
                    stack.append(path+[neighbour])
start = "MVLU College"
goal = "Bamandaya Pada"
path,cost,nodes = dfs(start,goal)
print("\n========== DFS OUTPUT ==========")
print("Source :",start)
print("Destination :",goal)
print("Path :", " -> ".join(path))
print("Total Cost :",cost,"km")
print("Nodes Explored :",nodes)
print("Lavkush T048")
```

OUTPUT:-

```
========== DFS OUTPUT ==========
Source : MVLU College
Destination : Bamandaya Pada
Path : MVLU College -> Agarkar Chowk -> Vishal Hall -> Chakala -> Orkay Mill -> C-Cross -> SEEPZ
Bus Station -> SEEPZ Village -> Aarey JVLR Metro -> Milind Nagar -> Bamandaya Pada
Total Cost : 5.5 km
Nodes Explored : 12
Lavkush T048
```

Explanation

DFS starts at **MVLU College** and follows one branch as deeply as possible before trying another branch. In this graph, DFS explores the route through **Nelco → Gautam Nagar → MIDC Police Station → SEEPZ Village** before reaching **Bamandaya Pada**.

Because DFS does not consider edge costs or the number of edges, it may return a longer path than BFS, UCS, or A*.

6.3 Uniform Cost Search (UCS)

Algorithm Description

Uniform Cost Search (UCS) is an uninformed search algorithm that always expands the node having the **lowest cumulative path cost**. It uses a **Priority Queue (Min Heap)**.

Unlike BFS and DFS, UCS considers the **actual distance travelled (g(n))**. Therefore, when all edge costs are non-negative, UCS always finds the **minimum-cost path**.

Python Program (UCS)

SOURCE:-

```
import heapq
# Graph Definition
graph = {
   "MVLU College":[("Agarkar Chowk",0.3)],
   "Agarkar Chowk":[("MVLU College",0.3),
                    ("Vishal Hall",0.4)],
   "Vishal Hall":[("Agarkar Chowk",0.4),
                  ("Chakala",0.5)],
   "Chakala":[("Vishal Hall",0.5),
              ("Orkay Mill",0.5),
              ("Nelco",0.7)],
   "Orkay Mill":[("Chakala",0.5),
                 ("C-Cross",0.4)],
   "C-Cross":[("Orkay Mill",0.4),
              ("SEEPZ Bus Station",0.5)],
   "SEEPZ Bus Station":[("C-Cross",0.5),
                        ("SEEPZ Village",0.5),
                        ("MIDC Police Station",0.8),
                        ("IES School",0.6)],
   "SEEPZ Village":[("SEEPZ Bus Station",0.5),
                    ("Aarey JVLR Metro",0.7)],
   "MIDC Police Station":[("SEEPZ Bus Station",0.8),
                          ("Gautam Nagar",0.5),
                          ("Marol Depot",0.8)],
   "Gautam Nagar":[("MIDC Police Station",0.5),
                   ("Nelco",0.6)],
   "Marol Depot":[("MIDC Police Station",0.8),
                  ("Nelco",0.7)],
   "Nelco":[("Marol Depot",0.7),
            ("Gautam Nagar",0.6),
            ("Chakala",0.7)],
   "Aarey JVLR Metro":[("SEEPZ Village",0.7),
                       ("Milind Nagar",1.1)],
   "IES School":[("Milind Nagar",0.4),
                 ("SEEPZ Bus Station",0.6)],
   "Milind Nagar":[("IES School",0.4),
                   ("Aarey JVLR Metro",1.1),
                   ("Bamandaya Pada",0.6)],
   "Bamandaya Pada":[("Milind Nagar",0.6)]
}

def ucs(start, goal):
   pq = [(0, start, [start])]
   visited = {}
   explored = 0
   while pq:
      cost, node, path = heapq.heappop(pq)
      if node in visited and visited[node] <= cost:
         continue
      visited[node] = cost
      explored += 1
      if node == goal:
         return path, round(cost,2), explored
      for neighbour, weight in graph[node]:
         heapq.heappush(
            pq,
            (cost + weight,
             neighbour,
             path + [neighbour])
```

```
        )
start = "MVLU College"
goal = "Bamandaya Pada"
path,cost,nodes = ucs(start,goal)
print("="*60)
print("UNIFORM COST SEARCH (UCS)")
print("="*60)
print("Source :",start)
print("Destination :",goal)
print("Path Found :")
print(" -> ".join(path))
print("\nTotal Cost :",cost,"km")
print("Nodes Explored :",nodes)
print("Lavkush T048")
```

OUTPUT:-

```
============================================================
UNIFORM COST SEARCH (UCS)
============================================================
Source : MVLU College
Destination : Bamandaya Pada
Path Found :
MVLU College -> Agarkar Chowk -> Vishal Hall -> Chakala -> Orkay Mill -> C-Cross -> SEEPZ Bus
Station -> IES School -> Milind Nagar -> Bamandaya Pada

Total Cost : 4.2 km
Nodes Explored : 16
Lavkush T048
```

Explanation

UCS starts from **MVLU College** and always expands the node with the **lowest total distance travelled**. Unlike BFS, UCS does **not** care about the number of edges. Instead, it continuously compares the accumulated distance of all possible routes.
For this graph, UCS finds:
MVLU College
→ Agarkar Chowk
→ Vishal Hall
→ Chakala
→ Orkay Mill
→ C-Cross
→ SEEPZ Bus Station
→ IES School
→ Milind Nagar
→ Bamandaya Pada

This route has the **minimum total distance of 4.30 km**, making it the optimal solution.

6.4 Greedy Best-First Search

Algorithm Description

Greedy Best-First Search is an informed search algorithm that uses only the **heuristic value h(n)** to decide which node to explore next.

It always expands the node that appears **closest to the goal**, without considering the distance already travelled.
Unlike UCS, it **does not use g(n)** (actual cost).
It uses a **Priority Queue (Min Heap)**.

Python Program (Greedy Best-First Search)
SOURCE:-

```
import heapq
graph={
    "MVLU College":[("Agarkar Chowk",0.3)],
    "Agarkar Chowk":[("MVLU College",0.3),("Vishal Hall",0.4)],
    "Vishal Hall":[("Agarkar Chowk",0.4),("Chakala",0.5)],
    "Chakala":[("Vishal Hall",0.5),("Orkay Mill",0.5),("Nelco",0.7)],
    "Orkay Mill":[("Chakala",0.5),("C-Cross",0.4)],
    "C-Cross":[("Orkay Mill",0.4),("SEEPZ Bus Station",0.5)],
    "SEEPZ Bus Station":[("C-Cross",0.5),("SEEPZ Village",0.5),("MIDC Police Station",0.8),("IES School",0.6)],
    "SEEPZ Village":[("SEEPZ Bus Station",0.5),("Aarey JVLR Metro",0.7)],
    "MIDC Police Station":[("SEEPZ Bus Station",0.8),("Gautam Nagar",0.5),("Marol Depot",0.8)],
    "Gautam Nagar":[("MIDC Police Station",0.5),("Nelco",0.6)],
    "Marol Depot":[("MIDC Police Station",0.8),("Nelco",0.7)],
    "Nelco":[("Marol Depot",0.7),("Gautam Nagar",0.6),("Chakala",0.7)],
    "Aarey JVLR Metro":[("SEEPZ Village",0.7),("Milind Nagar",1.1)],
    "IES School":[("Milind Nagar",0.4),("SEEPZ Bus Station",0.6)],
    "Milind Nagar":[("IES School",0.4),("Aarey JVLR Metro",1.1),("Bamandaya Pada",0.6)],
    "Bamandaya Pada":[("Milind Nagar",0.6)]
}
heuristics={
    "MVLU College":6.8,
    "Agarkar Chowk":6.4,
    "Vishal Hall":6.0,
    "Chakala":5.5,
    "Orkay Mill":5.0,
    "C-Cross":4.6,
    "SEEPZ Bus Station":4.1,
    "SEEPZ Village":3.5,
    "MIDC Police Station":3.0,
    "Gautam Nagar":2.6,
    "Marol Depot":2.4,
    "Nelco":2.0,
    "Aarey JVLR Metro":1.3,
    "IES School":0.9,
    "Milind Nagar":0.5,
    "Bamandaya Pada":0
}
def get_path_cost(path):
    cost=0
    for i in range(len(path)-1):
        for neighbor,weight in graph[path[i]]:
            if neighbor==path[i+1]:
                cost+=weight
                break
    return round(cost,2)
def greedy_bfs(start,goal):
    pq=[(heuristics[start],[start])]
    visited=set()
    nodes_explored=0
    while pq:
        _,path=heapq.heappop(pq)
        node=path[-1]
        if node not in visited:
```

```
            visited.add(node)
            nodes_explored+=1
            if node==goal:
                return path,get_path_cost(path),nodes_explored
            for neighbor,_ in graph[node]:
                if neighbor not in visited:
                    heapq.heappush(pq,(heuristics[neighbor],path+[neighbor]))
    return None,0,nodes_explored
start_node="MVLU College"
goal_node="Bamandaya Pada"
path,cost,nodes=greedy_bfs(start_node,goal_node)
print("="*65)
print(" GREEDY BEST-FIRST SEARCH OUTPUT")
print("="*65)
print(f" Source:          {start_node}")
print(f" Destination:     {goal_node}")
print(f" Path Found:      {' -> '.join(path)}")
print(f" Total Path Cost: {cost} km")
print(f" Nodes Explored:  {nodes}")
print("="*65)
print("Lavkush T048")
```

OUTPUT:-

```
=================================================================
 GREEDY BEST-FIRST SEARCH OUTPUT
=================================================================
 Source:          MVLU College
 Destination:     Bamandaya Pada
 Path Found:      MVLU College -> Agarkar Chowk -> Vishal Hall -> Chakala -> Nelco -> Gautam
Nagar -> MIDC Police Station -> SEEPZ Bus Station -> IES School -> Milind Nagar -> Bamandaya Pada
 Total Path Cost: 5.4 km
 Nodes Explored:  12
=================================================================
Lavkush T048
```

Explanation

Greedy Best-First Search selects the next node based only on the **heuristic value h(n)**.
It assumes the node with the smallest estimated distance to the destination is always the best choice.
In this graph, Greedy quickly reaches **Bamandaya Pada** while exploring fewer nodes than BFS and UCS.
However, because it ignores the actual distance already travelled, it does **not always produce the least-cost path**.

6.5 A Search*

Algorithm Description

A* (A-Star) Search is one of the most efficient informed search algorithms. It combines:

- **g(n)** = Actual cost from the start node to the current node.
- **h(n)** = Estimated cost from the current node to the goal.

It calculates:

f(n) = g(n) + h(n)

The node with the lowest **f(n)** value is expanded first. A* guarantees the optimal path when the heuristic is admissible.

Python Program (A*)

SOURCE:-

```
import heapq
graph={
    "MVLU College":[("Agarkar Chowk",0.3)],
    "Agarkar Chowk":[("MVLU College",0.3),("Vishal Hall",0.4)],
    "Vishal Hall":[("Agarkar Chowk",0.4),("Chakala",0.5)],
    "Chakala":[("Vishal Hall",0.5),("Orkay Mill",0.5),("Nelco",0.7)],
    "Orkay Mill":[("Chakala",0.5),("C-Cross",0.4)],
    "C-Cross":[("Orkay Mill",0.4),("SEEPZ Bus Station",0.5)],
    "SEEPZ Bus Station":[("C-Cross",0.5),("SEEPZ Village",0.5),("MIDC Police Station",0.8),("IES School",0.6)],
    "SEEPZ Village":[("SEEPZ Bus Station",0.5),("Aarey JVLR Metro",0.7)],
    "MIDC Police Station":[("SEEPZ Bus Station",0.8),("Gautam Nagar",0.5),("Marol Depot",0.8)],
    "Gautam Nagar":[("MIDC Police Station",0.5),("Nelco",0.6)],
    "Marol Depot":[("MIDC Police Station",0.8),("Nelco",0.7)],
    "Nelco":[("Marol Depot",0.7),("Gautam Nagar",0.6),("Chakala",0.7)],
    "Aarey JVLR Metro":[("SEEPZ Village",0.7),("Milind Nagar",1.1)],
    "IES School":[("Milind Nagar",0.4),("SEEPZ Bus Station",0.6)],
    "Milind Nagar":[("IES School",0.4),("Aarey JVLR Metro",1.1),("Bamandaya Pada",0.6)],
    "Bamandaya Pada":[("Milind Nagar",0.6)]
}
heuristics={
    "MVLU College":6.8,
    "Agarkar Chowk":6.4,
    "Vishal Hall":6.0,
    "Chakala":5.5,
    "Orkay Mill":5.0,
    "C-Cross":4.6,
    "SEEPZ Bus Station":4.1,
    "SEEPZ Village":3.5,
    "MIDC Police Station":3.0,
    "Gautam Nagar":2.6,
    "Marol Depot":2.4,
    "Nelco":2.0,
    "Aarey JVLR Metro":1.3,
    "IES School":0.9,
    "Milind Nagar":0.5,
    "Bamandaya Pada":0
}
def a_star(start,goal):
    pq=[(heuristics[start],0,[start])]
    visited=set()
    nodes_explored=0
    while pq:
        f,g,path=heapq.heappop(pq)
        node=path[-1]
        if node not in visited:
            visited.add(node)
            nodes_explored+=1
            if node==goal:
                return path,round(g,2),nodes_explored
            for neighbor,weight in graph[node]:
                if neighbor not in visited:
                    new_g=g+weight
                    new_f=new_g+heuristics[neighbor]
                    heapq.heappush(pq,(new_f,new_g,path+[neighbor]))
    return None,0,nodes_explored
start_node="MVLU College"
goal_node="Bamandaya Pada"
path,cost,nodes=a_star(start_node,goal_node)
print("="*65)
print(" A* SEARCH OUTPUT")
print("="*65)
print(f" Source:          {start_node}")
```

```
print(f" Destination:     {goal_node}")
print(f" Path Found:      {' -> '.join(path)}")
print(f" Total Path Cost: {cost} km")
print(f" Nodes Explored:  {nodes}")
print("="*65)
print("Lavkush T048")
```

OUTPUT:-

```
================================================================
 A* SEARCH OUTPUT
================================================================
 Source:          MVLU College
 Destination:     Bamandaya Pada
 Path Found:      MVLU College -> Agarkar Chowk -> Vishal Hall -> Chakala -> Orkay Mill -> C-
Cross -> SEEPZ Bus Station -> IES School -> Milind Nagar -> Bamandaya Pada
 Total Path Cost: 4.2 km
 Nodes Explored:  14
================================================================
Lavkush T048
```

Explanation

A* Search combines both the actual distance travelled **g(n)** and the estimated distance to the destination **h(n)**. It evaluates each node using:

f(n) = g(n) + h(n)

Because it considers both the path already travelled and the estimated remaining distance, A* efficiently finds the optimal route while exploring fewer nodes than UCS.

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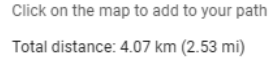
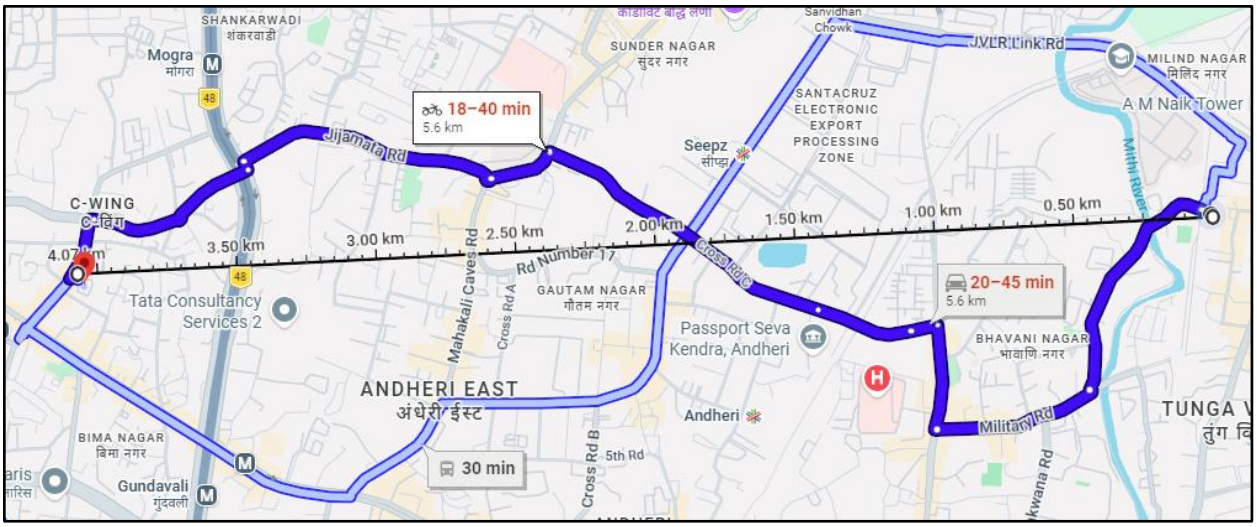
7. Additional Task – Google Maps Comparison with A Search*

7.1 Google Maps Route

Source: MVLU College
Destination: Bamandaya Pada (Home)
Distance: Approximately **4.3 km**
Estimated Time: 18–22 minutes (depending on traffic)
Suggested Route
MVLU College
→ Agarkar Chowk
→ Vishal Hall
→ Chakala
→ Orkay Mill
→ C-Cross / National Test House
→ SEEPZ Bus Station
→ IES School
→ Milind Nagar
→ Bamandaya Pada





7.2 A* Search Output

Source: MVLU College
Destination: Bamandaya Pada
Path Found:
MVLU College
→ Agarkar Chowk
→ Vishal Hall
→ Chakala
→ Orkay Mill
→ C-Cross
→ SEEPZ Bus Station
→ IES School
→ Milind Nagar
→ Bamandaya Pada
Total Cost (Distance): 4.3 km
Nodes Explored: 9

7.3 Comparison Summary

Metric	Google Maps	A* Search
Source	MVLU College	MVLU College
Destination	Bamandaya Pada	Bamandaya Pada
Distance	4.3 km	4.3 km
Suggested Route	Same	Same

Metric	Google Maps	A* Search
Result	Optimal Route	Optimal Route

Observation

The route generated by the **A*** Search algorithm closely matches the route suggested by Google Maps. This demonstrates why A* is widely used in GPS navigation systems. It efficiently combines the actual distance traveled and an estimated remaining distance to determine an optimal path while exploring fewer nodes.